

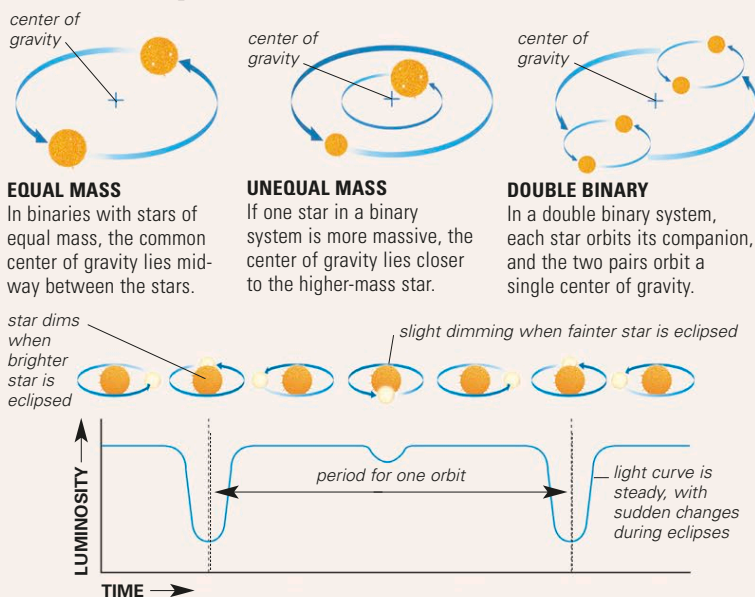
MULTIPLE STARS

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A **MULTIPLE STAR** IS A **SYSTEM** of two or more stars bound together by gravity. Systems with two stars are called **binary** or **double stars**. Although at first sight only a few stars appear to be multiple, it is estimated that they may account for over 60 percent of stars in the Milky Way. Binary stars orbit each other at a great variety of distances, with orbital periods ranging from a few hours to millions of years. Multiple stars allow astronomers to determine stellar masses and diameters and give them insights into stellar evolution.

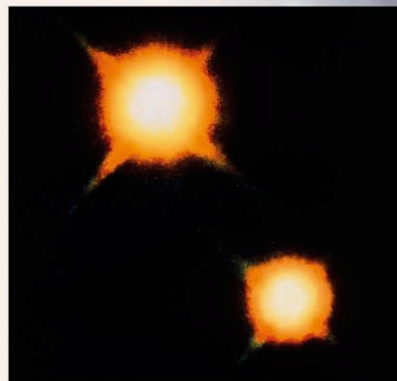
BINARIES AND BEYOND

Although there are many millions of multiple systems within the Milky Way, not all of them consist of just two stars in mutual orbit. What may appear to be a double or binary star can often reveal itself to be a more complex system of three or more stars. A simple binary system consists of two stars orbiting each other. If the stars are of similar mass, they orbit around a common center of gravity, located between them. If one of the stars is much more massive than the other, the common center of gravity may be located inside the massive star. The more massive star then merely exhibits a wobble, while the secondary star appears to take on all the orbital motion. However, multiple systems may have a greater orbital complexity, with multiple centers of gravity. For example, a quadruple system may have two pairs of stars orbiting each other, while the individual stars within the pair are also in mutual orbit.



BRIGHT BINARY

One of the brightest stars in our sky, Alpha (α) Centauri is also a striking double star, with two sunlike components that orbit each other in just under 80 years.



DETECTING BINARIES

Astronomers detect binary stars in a variety of ways. Line-of-sight binaries consist of stars that appear in the sky to be related but, in fact, are not physically associated. These are usually identified by determining the distances to the individual stars. Visual binaries are detected when the naked eye or magnification splits the stars and shows each one separately. Measurements of each star's position over time allow astronomers to compute their orbit. Although they cannot be separated with a telescope, astrometric binaries are detected when an unseen companion causes a star to wobble periodically through its gravitational influence. Spectroscopy can also be used to identify binary stars, when a star's spectrum appears doubled up and actually consists of the combined

ECLIPSING BINARIES

Eclipsing binary stars are detected by variations in a star's magnitude. These variations occur when stars periodically pass in front of each other during orbit.

spectrums of two stars orbiting each other. These systems are known as spectroscopic binaries. The apparent magnitude of a binary star may show periodic fluctuations, caused by the stars eclipsing each other. Such stars are known as eclipsing binaries.

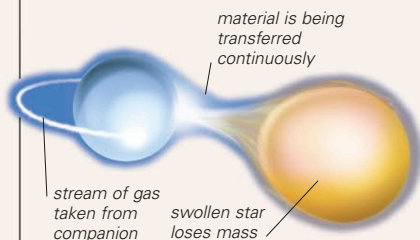
CO-EVOLUTION

Like all stars, those within a multiple system evolve. A binary system can start out as two main-sequence stars with a mutual, regular orbit and predictable eclipses. However, over millions of years, the stars progress through their evolutionary stages, which may result in a binary system with two stars of completely different characteristics. One example is the Sirius system (see p.252). The evolution of one star within a system can change the behavior of the whole system. For example, should a star expand and become a red giant, the expansion can bring the evolving star to interact with its companion star. This leads to mass transfer, and if the companion has itself evolved into a white

dwarf, the result can be a cataclysmic explosion (see p.283). Stellar evolution can thus convert a stable binary system into a scene of immense violence.

INTERACTING BINARIES

The stars in some binary systems are located so close together that material can pass between them. Here, one of the stars has swollen and is spilling gas onto the other.



EXTREME BINARIES

Many binary systems exhibit perfectly regular behavior, with the stars orbiting each other for millions of years with no dramatic changes. However, other binary systems, particularly those that have undergone evolutionary changes, may exhibit much more extreme behavior. One example is a contact binary system, in which the two stars are touching each other. In this case, a massive star transfers material to the secondary star at a faster rate than the secondary star can absorb. This results in the material forming a common envelope that surrounds both stars. The envelope then creates frictional drag, causing the stars' orbital periods to change. In this way, a binary system with a wide separation and an orbital period of about a decade may be converted into a rapid system with the stars orbiting in a matter of hours. Other binary systems seem to operate at the extremes of physics. The discovery in 1974 of a binary pulsar system opened up a new field of observation in gravitational physics. A strong source of gravitational waves, binary pulsars are very regular and precise systems.